User Guide - Direct method.

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Suppose we have given a symmetric shift register with parameters k, p and n. The goal is to determine minimal periods in the general case. This will be done by a direct method based on the definition of the given symmetric shift register. This note also describes notation and assumptions.

The direct method has certain limitations. The program stops and give no answer if the minimal period is larger than 10 000 000. This method also works for longer periods, but we had to set a limit. The reason a limit must be set is that the direct method is computationally intensive. However, if a higher limit is desired, it is also possible to change this limit. But 10 000 000 is the default value.

Strings generated from vectors are defined in Section 4. Moreover, the note "The direct method" gives a brief summary of the method being used.

1. Symmetric shift registers.

Let k, p and n be integers such that $0 \leq k \leq k+p < n$. The symmetric shift register $\theta : \{0,1\}^n \to \{0,1\}^n$ with respect to k, p and n is defined by $\theta(a_1 \cdots a_n) = a_2 \cdots a_{n+1}$ where

$$a_{n+1} = a_1' \text{ if } k \leq a_2 + \cdots + a_n \leq k+p, \text{ and } a_{n+1} = a_1 \text{ otherwise,}$$

where $a_1' = 0$ if $a_1 = 1$, and $a_1' = 1$ if $a_1 = 0$.

We let $A^\infty = a_1 \cdots a_n a_{n+1} a_{n+2} \cdots$ be the string generated from A by the symmetric shift register with parameters k, p and n. For $i \geq 0$, then

$a_{n+i+1} = a_{i+1}'$ if $k \leq a_{i+2} + \cdots + a_{i+n} \leq k+p$. Otherwise, $a_{n+i+1} = a_{i+1}$.

The minimal period of A^∞ is called the minimal period of A with respect to the symmetric shift register with parameters k, p and n.

2. Input.

A vector $Q = (q_1, \cdots, q_s)$ satisfying

$$s \text{ is even, } q_1 \geq 0,\ q_i > 0 \text{ for } 2 \leq i < s, \text{ and } q_s \geq 0,$$

must be entered into the first input field. The input vector should be comma separated and must be given without parenthesis. $n = q_1 + q_2 + \cdots + q_s$ is calculated. In the next input field the value of p should be entered. The p-value must satisfy $0 \leq p < n$.

In the third input field the value of k is entered. The k-value must satisfy

$$k_{min} \leq k \leq k_{max} \tag{2.1}$$

where k_{min} and k_{max} are calculated by the program. In fact,

$$k_{min} = max\{0, w - (p+1\},$$
$$k_{max} = min\{w, n - (p+1)\},$$

where $w = q_1 + q_3 + \cdots + q_{s-1}$.

3. Output.

The program calculates the string $A = a_1 \cdots a_n$ corresponding to Q. The condition (2.1) ensures that

$0 \leq k \leq w(A) \leq k + p + 1 \leq n$ where $w(A)$ is the weight of A.

Suppose $A^\infty = a_1 \cdots a_n a_{n+1} a_{n+2} \cdots$ is generated from $A = a_1 \cdots a_n$ by the symmetric shift register with parameters k, p and n.

The minimal period of A with respect to the symmetric shift register with parameters k, p and n, will be calculated and displayed as output.

4. Vector representations.

Suppose $Q = (q_1, \cdots, q_s)$ where s is even, $q_1 \geq 0$, $q_i > 0$ for $1 \leq i < s$, and $q_s \geq 0$. Then Q is the even vector representation of the string

$$A = 1_{q_1} 0_{q_2} 1_{q_3} 0_{q_4} \cdots 1_{q_{s-1}} 0_{q_s}.$$

Example 4.1. If $Q = (2, 3, 4, 1, 2, 2)$, then Q represents
$A = 1_2 0_3 1_4 0_1 1_2 0_2 = 11000111101100$.

Example 4.2. If $Q = (3, 4, 2, 0)$, then Q represents
$A = 1_3 0_4 1_2 0_0 = 111000011$.

Example 4.3. If $Q = (2, 3, 1, 3)$, then Q represents
$A = 1_2 0_3 1_1 0_3 = 110001000$.

Example 4.4. If $Q = (0, 3, 1, 0)$, then Q represents $A = 1_0 0_3 1_1 0_0 = 0001$.

Example 4.5. If $Q = (0, 3, 1, 5)$, then Q represents
$A = 1_0 0_3 1_1 0_5 = 000100000$.